

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Voice based communication understanding analyzer
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely access the application. (<i>Current version supports local access; user login can be added in future.</i>)	Medium
2	Authorization Levels	The system supports user roles such as Student and Educator for accessing evaluation features and reports.	Medium
3	External Interfaces	Integrates OpenAI Whisper for speech-to-text, Sentence-BERT for semantic similarity, Librosa for audio analysis, and ReportLab for PDF generation.	High
4	Transaction Processing	The system accepts audio files, processes speech, evaluates conceptual understanding, calculates scores, and generates reports.	High
5	Reporting	Generates an evaluation dashboard and downloadable PDF reports containing transcripts, semantic similarity scores, waveform images, speech metrics, and feedback.	High
6	Business Rules	The uploaded audio is transcribed, compared with reference content, evaluated using semantic similarity and speech quality metrics, and assigned a final understanding	High

S.No	Requirement Category	Requirement Description	Priority
		score with qualitative feedback.	
7	Compliance to Laws or Regulations	User audio and evaluation data should be handled securely and used only for educational purposes while maintaining user privacy.	Medium
8	Other	The system supports WAV and MP3 audio uploads, waveform visualization, and AI-generated performance feedback.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process uploaded audio and display results quickly.	Average response time < 5 seconds for standard audio files.
2	Scalability	The system should support additional users and larger datasets through future cloud deployment.	Supports 100+ users with scalable cloud architecture.
3	Security & Data Privacy	Uploaded audio files and generated reports should be protected from unauthorized access.	Secure local storage with API keys managed using .env files.
4	Reliability & Availability	The application should execute without failures during normal operation.	99% successful execution during testing.
5	Usability & Accessibility	The interface should be simple, responsive, and easy for students and educators to use.	Interactive Streamlit interface with minimal user training required.
6	Maintainability & Portability	The code should be modular, reusable, and executable on different operating systems.	Runs on Windows, Linux, and macOS using Python 3.10+ with modular Python files.